Missing Middle-Zoning in Toronto: Affordability Under Alternative Policies

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1 Project Scope and Final Workflow

The project scope remains unchanged from Milestone 1. The simulation models Census Tract (CT) agents across Toronto to evaluate how alternative missing-middle zoning policies affect housing affordability over a 10-year horizon, subject to infrastructure capacity constraints and an uncertain market response. The City of Toronto serves as the primary stakeholder.

Revision from Milestone 1 feedback: The simulation time scale has been refined from annual to quarterly steps (T=40 steps over 10 years) to allow for more gradual accumulation of market effects and greater differentiation between scenarios. Demand, development probability, and infrastructure growth parameters have been scaled accordingly to preserve annual equivalence.

The final workflow is presented in Figure 1. City-level housing demand growth is treated as exogenous, while development, prices, rent, vacancy, and infrastructure strain evolve endogenously within each CT agent. The Policy Model defines zoning eligibility and incentive parameters by scenario, which feed into the Demand Allocation Model. The Development Model translates demand pressures and infrastructure constraints into probabilistic changes in housing supply within each CT, calibrated against observed 2016–2021 CT-level growth using a two-stage ML model. Density feeds back into the Infrastructure Model, creating a strain feedback loop. The primary output metric calculated at each time step is the Affordability Index.

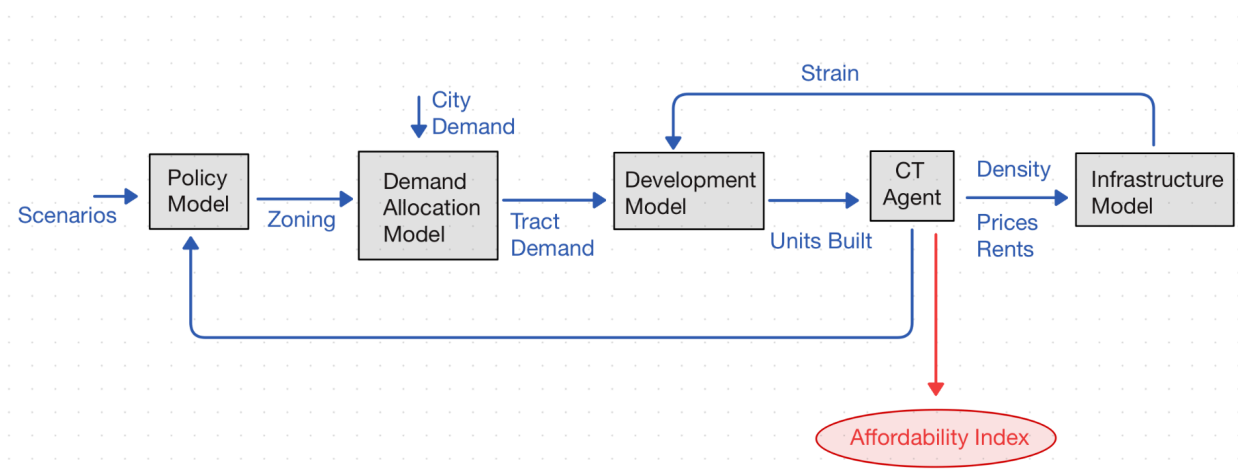

Figure 1: Agent-based Workflow for Project

2 Agent's Attributes

2.1 Census Tract Agent Attributes

Table 1: Census Tract (CT) Agent Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Median household income [$/yr]	Statistics Canada, Census Profile 2021, Cat. 98-401-X2021007, Characteristic ID 243	Direct value; see 1) below
Number of households [count]	Statistics Canada, Census Profile 2021, Characteristic ID 50	Direct value
Total private dwellings [units]	Statistics Canada, Census Profile 2021, Characteristic ID 4	Direct value; initialized as units_total
Renter households [count]	Statistics Canada, Census Profile 2021, Characteristic ID 1416	Direct value; used to derive renter_share
Owner households [count]	Statistics Canada, Census Profile 2021, Characteristic ID 1415	Direct value
Renter share [fraction]	Derived from Census Profile 2021	renter_share = renter_households / total_households_by_tenure; see 2) below
Home price [$]	Statistics Canada, Census Profile 2021, Characteristic ID 1488	2021, Characteristic ID 1488Median dwelling value used as proxy for ownership cost; see 3) below
Annual rent [$/yr]	Statistics Canada, Census Profile 2021, Characteristic ID 1494	Median monthly shelter cost for renters × 12; see 4) below
Vacancy rate [fraction]	Canada Mortgage and Housing Corporation (CMHC), Housing Market Outlook, Toronto CMA, 2021	Initialized at 0.05 for all CTs (Toronto average); no CT-level vacancy data available from census; see 5) below
Demand pressure [index]	Derived during simulation	Allocated by

		DemandAllocationModel each time step; initialized at 0
Infrastructure strain [index]	Derived during simulation	Computed as infra_load / infra_capacity; initialized at 0
Transit indicator [0–1]	City of Toronto Open Data, TTC Routes and Schedules (GTFS)	Normalized count of rapid transit stops within 800m of CT centroid; see 6) below
Population [count]	Statistics Canada, Census Profile 2021, Characteristic ID 1	Direct value; used as a density proxy in ML features

Notes:

1. Median total household income in 2020, reported at the CT level. All 1,220 Toronto CTs with non-suppressed income data were retained. This is the primary driver of the Affordability Index output metric [1].
2. Renter share is derived as renter_households / total_households_by_tenure (Characteristic IDs 1416 and 1414, respectively). This attribute is correlated with income: lower-income CTs tend to have higher renter shares ($r \approx -0.45$ in the Toronto sample) [2]. This correlation is not explicitly modelled but is captured implicitly through the ML calibration.
3. Median dwelling value is used as a proxy for the housing cost. This understates true transaction prices for new buyers since it reflects existing stock, including long-held properties. 8 CTs with suppressed dwelling values were imputed using the Toronto CT median ($955,000), which is documented as a limitation [3].
4. Annual rent is derived as median monthly shelter cost for renters × 12 (Characteristic ID 1494). This reflects costs for existing tenants, many of whom are in rent-controlled units, and therefore understates new market rents [4]. This is acknowledged as a limitation affecting the initial calibration of AI_rent.
5. No CT-level vacancy rate is available from the census. The CMHC reports a Toronto CMA average vacancy rate of approximately 3–5% for 2021 [5]. We initialize all CTs at 0.05 and allow vacancy to evolve endogenously through the market update model. This is a simplification that will be revisited in the Final Report.
6. The transit indicator is computed as the count of rapid transit stops (subway and streetcar) within 800m of each CT centroid, normalized by the maximum value across all Toronto CTs. CT centroids were computed from Statistics Canada 2021 CT boundary files (Catalogue 92-160-X), projected to UTM Zone 17N (EPSG:32617) for accurate metre-based distance calculations [6]. The 500m threshold for binary eligibility under Scenario 2 follows City of Toronto Official Plan guidelines for transit-oriented development planning areas [7]. A limitation is that high-frequency bus routes are excluded from the definition of rapid transit.

2.2 Policy Model Attributes

Table 2: Policy Model (Pseudo-Agent) Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Zoning eligibility [binary]	Defined by scenario	S0: no CTs eligible; S1: all 1,220 CTs eligible; S2: 181 CTs within 500m of rapid transit; S3: all 1,220 CTs eligible
Incentive level [0–1]	Assumption	S0–S2: 0.0; S3: 0.5, represents a 50% boost to development probability to simulate reduced development charges
Scenario type [categorical]	Defined by scenario	One of {S0, S1, S2, S3}

2.3 Demand Allocation Model Attributes

Table 3: Demand Allocation Model (Pseudo-Agent) Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Base demand [fraction of stock/quarter]	CMHC Housing Market Outlook, Toronto CMA	Set to 2% of the total housing stock per quarter; see 7) below
Demand growth [fraction/quarter]	Assumption based on CMHC projections	0.25%/quarter ≈ 1%/year household formation growth
Attractiveness weights [index]	Derived	1 / home_price per CT, normalized to sum to 1; cheaper CTs attract proportionally more demand

Notes:

7. Toronto added approximately 15,000 net new households per year between 2016 and 2021 (CMHC Housing Market Outlook) [8]. Expressed as a fraction of the total housing stock (~600,000 units), this is approximately 2.5% per year or 0.625% per quarter. We use 2% per quarter as a conservative estimate to account for the fact that not all demand translates to occupied new units within a single quarter.

2.4 Infrastructure Model Attributes

Table 4: Infrastructure Model (Pseudo-Agent) Attributes

Attribute [units]	**Data Source/Assumption**	**Input Model/Notes**
Infrastructure capacity [index]	Assumption	Initialized at 1.0 for all CTs; placeholder pending calibration in Final Report
Capacity growth rate [fraction/step]	Assumption	g_base = 0.01/quarter; reduced by incentive_level × 0.005 for S3 to reflect reduced development charge revenue
Load parameters (ω_0, ω_1)	Assumption	$\omega_0 = 0.5$, $\omega_1 = 0.3$; placeholder values; sensitivity analysis planned for Final Report

3 Agent's Behaviours

3.1 Census Tract Agent Behaviour

Table 5: Census Tract (CT) Agent Behaviour

Behaviour	**Description**
Receives allocated housing demand	demand_pressure updated by DemandAllocationModel each time step
Updates the housing supply when development occurs	apply_development() adds units split by tenure
Updates vacancy, prices, and rents	update_market() responds to the demand-supply gap
Accumulates infrastructure strain	update_infrastructure() computes load relative to capacity
Computes Affordability Index	affordability_own() and affordability_rent() at each time step

3.1.1 Receives Allocated Housing Demand

Demand pressure is allocated to each CT by the Demand Allocation Model at each time step and stored as `demand_pressure`. The allocation is described under the Demand Allocation Model behaviours below.

3.1.2 Updates Housing Supply When Development Occurs

When the Development Model determines that development occurs in a CT, new units are added and split between rental and ownership tenure as follows:

$$units_rent_added = round(units_added \times clip(renter_share,\ 0.2,\ 0.8))$$

$$units_own_added = units_added - units_rent_added$$

$$units_total \mathrel{+}= units_added$$

The renter share is clipped to [0.2, 0.8] to prevent degenerate outcomes in which all new units are assigned entirely to one tenure type. This constraint reflects the reality that mixed-tenure development is typical under missing-middle zoning forms such as duplexes and multiplexes.

3.1.3 Updates Vacancy, Prices, and Rents

At each time step, the CT agent updates its vacancy rate in response to the demand-supply balance, then updates prices and rents in response to the vacancy gap from a target rate:

$$d_per_unit = \frac{demand_pressure}{max(units_total,\ 1)}$$

$$vacancy_rate_{t+1} = clip(vacancy_rate_t - 0.02 + 0.5 \times d_per_unit,\ 0,\ 0.25)$$

$$home_price_{t+1} = home_price_t \times (1 + \kappa_p \times (v^* - vacancy_rate_{t+1}))$$

$$annual_rent_{t+1} = annual_rent_t \times (1 + \kappa_r \times (v^* - vacancy_rate_{t+1}))$$

where κ_p = 0.15 is the ownership price elasticity with respect to the vacancy gap, κ_r = 0.10 is the rental price elasticity, and v^* = 0.03 is the target vacancy rate based on the CMHC Toronto CMA average [9]. When vacancy exceeds the target, prices fall; when vacancy is below the target, prices rise. These elasticity parameters are placeholder values and will be calibrated against observed Toronto price trends in the sensitivity analysis for the Final Report.

3.1.4 Accumulates Infrastructure Strain

Infrastructure load and strain are updated each time step using a linear density proxy:

$$density_proxy = \frac{units_total}{max(households, 1)}$$

$$infra_load = \omega 0 + \omega 1 \times density_proxy$$

$$strain = \frac{infra_load}{max(infra_capacity, 10^{-6})}$$

where ω_0 = 0.5 and ω_1 = 0.3 are placeholder parameters. Infrastructure capacity grows each quarter by gcapg_{cap} gcap, which is reduced under Scenario 3 to reflect reduced municipal revenue from development charge incentives (see Infrastructure Model below).

3.1.5 Computes Affordability Index

The primary output metric is computed at each time step as:

$$AI^{own}{}_{ct,\,t} = \frac{median_income_{ct}}{home_price_t}, \; AI^{rent}{}_{ct,\,t} = \frac{median_income_{ct}}{annual_rent_{ct,\,t}}$$

Higher values indicate more affordable housing. Median income is treated as a static attribute since census income data reflects a point-in-time measure, and endogenous income dynamics are outside the scope of this model.

3.2 Policy Model Attributes

Table 6: Policy Model (Pseudo-Agent) Attributes

Behaviour	**Description**
Applies zoning rules across CTs	Sets eligible_ctuids based on scenario
Modifies development conditions by scenario	Sets incentive_level; used to boost p_dev and penalize infra capacity growth

The Policy Model behaviour 'Applies Zoning Rules' is represented as:

$$eligible_{S0}(ct) = 0$$

$$eligible_{\{S1,\,S3\}}(ct) = all\ CTs$$

$$eligible_{S2}(ct) = \{ct: near_rapid_500m = 1\}$$

Any CT where *eligible(ct)* = False returns zero units regardless of the Development Model output. Under Scenario 3, *incentive_level* = 0.5, which modifies both development probability (see Development Model) and infrastructure capacity growth (see Infrastructure Model).

3.3 Demand Allocation Model Attributes

Table 7: Demand Allocation Model (Pseudo-Agent) Attributes

Behaviour	**Description**
Distributes housing demand across CTs	Allocates city demand proportional to CT attractiveness
Updates demand pressure in each CT	Sets ct.demand_pressure each time step

The Demand Allocation Model behaviour 'Distributes Housing Demand' is represented as:

$$city_demand_t = base_demand \times (1 + g_d)^t \times \varepsilon_t, \quad \varepsilon_t \sim LogNormal(0,\ 0.1)$$

$$attractiveness_i = \frac{1}{max(home_price_i, 1)}$$

$$demand_pressure_i = city_demand_t \ x \ \frac{attractiveness_i}{\sum_j attractiveness_j}$$

where g_d = 0.0025 per quarter and ε_t is a log-normal shock capturing uncertainty in household formation. CTs with lower home prices receive proportionally more demand pressure, reflecting the tendency for households to seek more affordable areas [10]. This is a simplification; the Final Report will extend the attractiveness function to incorporate transit proximity and affordability-based weights.

3.4 Development Model Attributes

Table 8: Development Model (Pseudo-Agent) Attributes

Behaviour	**Description**
Evaluates whether development occurs	Two-stage ML model: Stage 1 predicts P(development)
Determines units added	Stage 2 predicts the magnitude, conditional on development occurring

The Development Model behaviour 'Evaluates Whether Development Occurs' uses a logistic regression classifier (Stage 1) fitted on observed 2016–2021 CT-level housing growth. The development probability for CT *i* at time step *t* is:

$$p_{dev} = LogisticRegression(x_{i,t})$$

$$p_{dev,\,adjusted} = min\,(1,\ \frac{p_{dev}}{T} \times (1 + 0.3 \times incentive_level))$$

$$dev_event_i \sim Bernoulli(p_{dev,\,adjusted})$$

where $x_{i,t}$ is the feature vector of CT attributes, T = 40 is the simulation horizon (used to scale the annual calibration probability to a quarterly probability), and the incentive adjustment applies only under Scenario 3. If *eligible(ct)* = False, $p_{dev,\,adjusted}$ = 0 regardless.

The behaviour 'Determines Units Added' (Stage 2) is represented as:

$$units_added = max(0,\ round(LinearRegression(x_{i,t})))$$

Stage 2 is applied only when dev_event_i = 1. Both models were calibrated on 1,078 Toronto CTs using StatCan census data (Stage 1 5-fold CV F1 = 0.73; Stage 2 5-fold CV R^2 = 0.31) [11]. The relatively low Stage 2 R^2 reflects heterogeneity across CTs and is acknowledged as a limitation. The linear model captures the mean development magnitude but explains limited variance. A more flexible model (e.g. random forest) will be evaluated in the Final Report.

3.5 Infrastructure Model Attributes

Table 8: Development Model (Pseudo-Agent) Attributes

Behaviour	**Description**
Updates infrastructure strain based on housing growth	Computes load and strain each time step
Feeds back into development constraints	Strain feeds into CT agent's update_infrastructure()

The Infrastructure behaviour 'Updates Infrastructure Capacity and Strain' is represented as:

$$g_{cap} = max(0,\ gbase - \lambda \times incentive_level)$$

$$infra_capacity_{t+1} = infra_capacity_t \times (1 + g_{cap})$$

Where g_{base} = 0.01 λ = 0.005. Under Scenario 3, the incentive level of 0.5 reduces quarterly capacity growth from 1% to 0.75%, reflecting reduced municipal development charge revenue. Infrastructure load and strain are then updated via the CT agent's `update_infrastructure()` method as described above.

4 Simulation

4.0 Algorithm

The simulation is implemented as a Monte Carlo loop over N = 100 realizations and T = 40 quarterly time steps (10-year horizon). The Python notebook submitted alongside this report contains a cell named **Simulation,** which includes the full loop. The organization of the simulation is as follows:

For each scenario $S \in \{S0, S1, S2, S3\}$, the following steps are executed:

1. Load 1,220 CT agents from processed census data and initialize all dynamic attributes to 2021 baseline values
2. Construct the Policy Model for scenario S
3. For each realisation $r \in \{1,...,N\}$:
    - Deep copy the agent list so each realization starts independently from the 2021 baseline
    - Initialize a Demand Allocation Model and Infrastructure Model with a realization-specific random seed (seed+rseed + r seed+r)
    - For each time step $t \in \{1,...,T\}$:
        1. **Demand allocation**: distribute city-wide demand across CTs proportional to attractiveness
        2. **Development**: for each CT, call `predict_development()` to sample whether development occurs and how many units are added; apply `apply_development()` if units > 0
        3. **Market update**: for each CT, call `update_market()` to update vacancy rate, home price, and annual rent
        4. **Infrastructure update**: for each CT, call `infra_model.step()` to update capacity and strain
        5. **Record outputs**: compute mean AI_own, AI_rent, total units, units added, mean home price, mean rent, and mean strain across all CTs; store in output arrays of shape (N, T)

The four scenarios are run in parallel using Python's `multiprocessing.Pool` with four workers, one per scenario, reducing total wall-clock time from approximately 24 minutes to 6–7 minutes on an 8-core machine.

4.1 Baseline Results (S0: Status Quo)

Figure 2 presents the distribution of ownership affordability trajectories across N = 100 realizations of the baseline scenario (S0), in which no missing-middle zoning reform is applied. Each light line represents one realization; the bold line shows the mean trajectory; the shaded bands show the 25th–75th and 5th–95th percentile ranges.

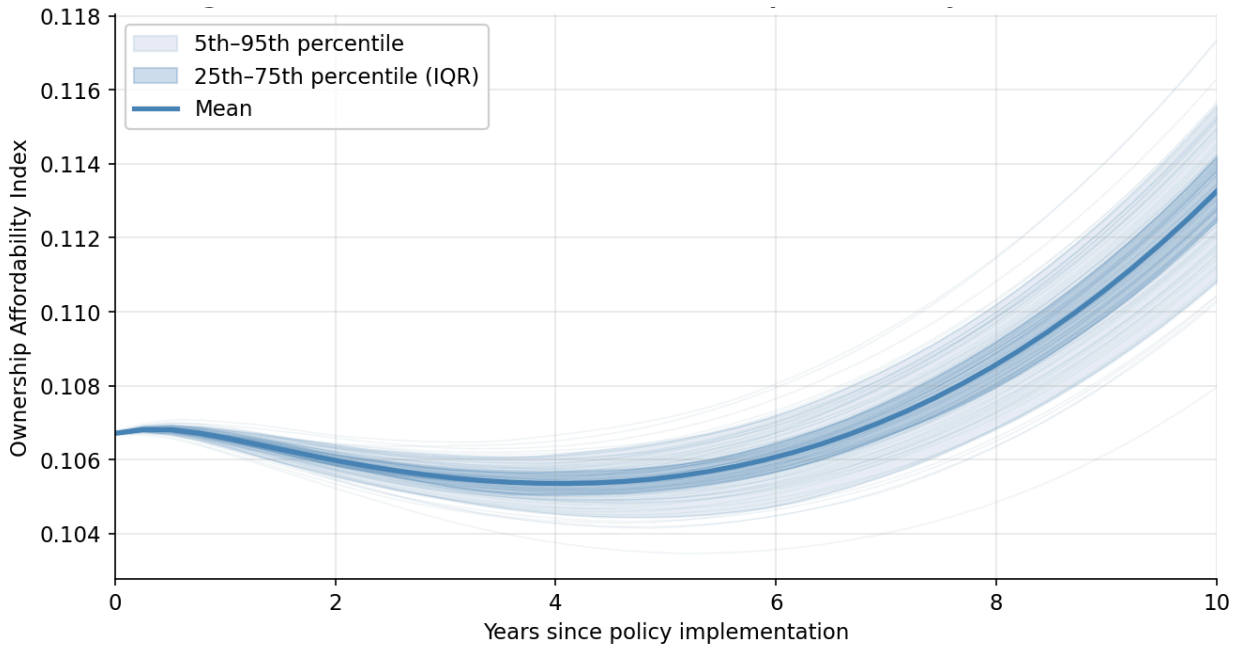

Figure 2: Baseline Scenario (S0): Ownership Affordability Over 10 Years

The baseline ownership affordability index begins at approximately 0.107 (median household income is roughly 10.7% of median home price, consistent with a price-to-income ratio of approximately 9× reported for Toronto by CMHC in 2021) and rises gradually to approximately 0.113 by year 10 [12]. The upward trend reflects the vacancy-price feedback mechanism: under the baseline, no new supply is added, and demand growth causes vacancy to fall below the target rate, which in turn drives prices upward, but income remains fixed, so the affordability index initially falls. Over longer periods, the model reaches a quasi-equilibrium where vacancy stabilises and prices grow more slowly, allowing the affordability index to recover. The IQR band is narrow, indicating low cross-realization variability in the baseline; the stochastic variation enters primarily through the log-normal demand shock.

Figure 2b presents the equivalent plot for rental affordability, which exhibits the same qualitative trajectory.

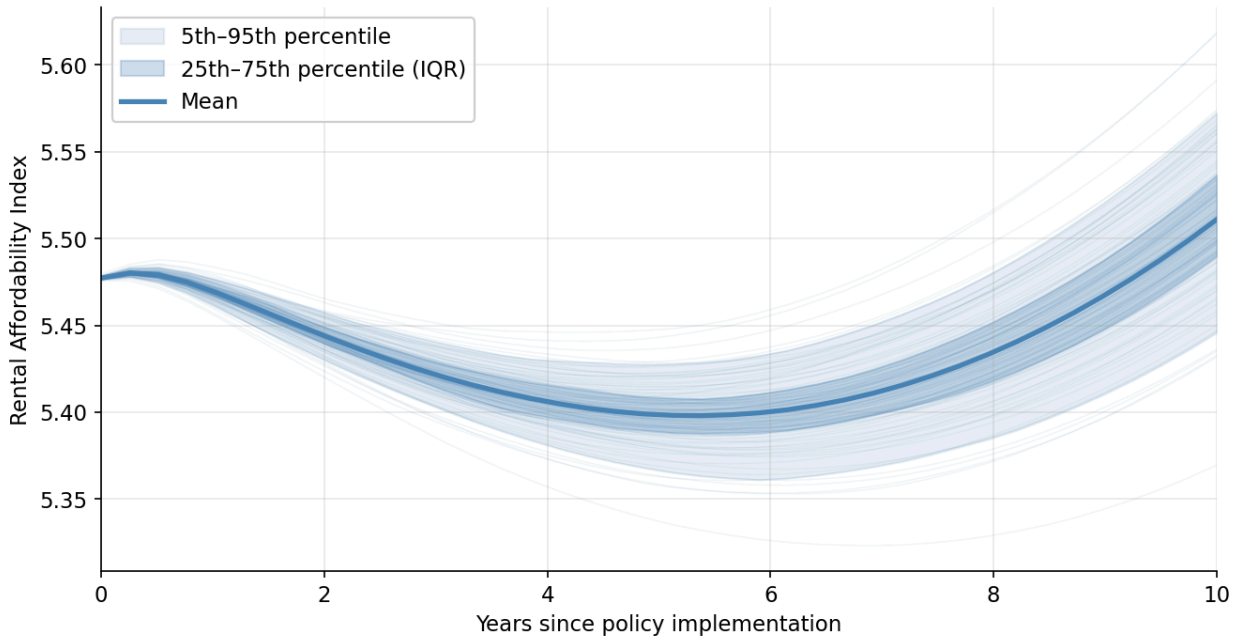

Figure 2b: Baseline Scenario (S0): Rental Affordability Over 10 Years

4.2 Baseline vs Scenario 1 Comparison

Figure 3 compares the distribution of ownership affordability at year 10 under S0 (status quo) and S1 (city-wide missing-middle zoning) across all 100 realizations, presented as a boxplot. A boxplot is used here rather than overlapping trajectory plots because it compactly conveys the median, interquartile range, and outliers for each scenario, allowing the stakeholder to directly compare the distribution of outcomes at the policy horizon without requiring interpretation of overlapping time series.

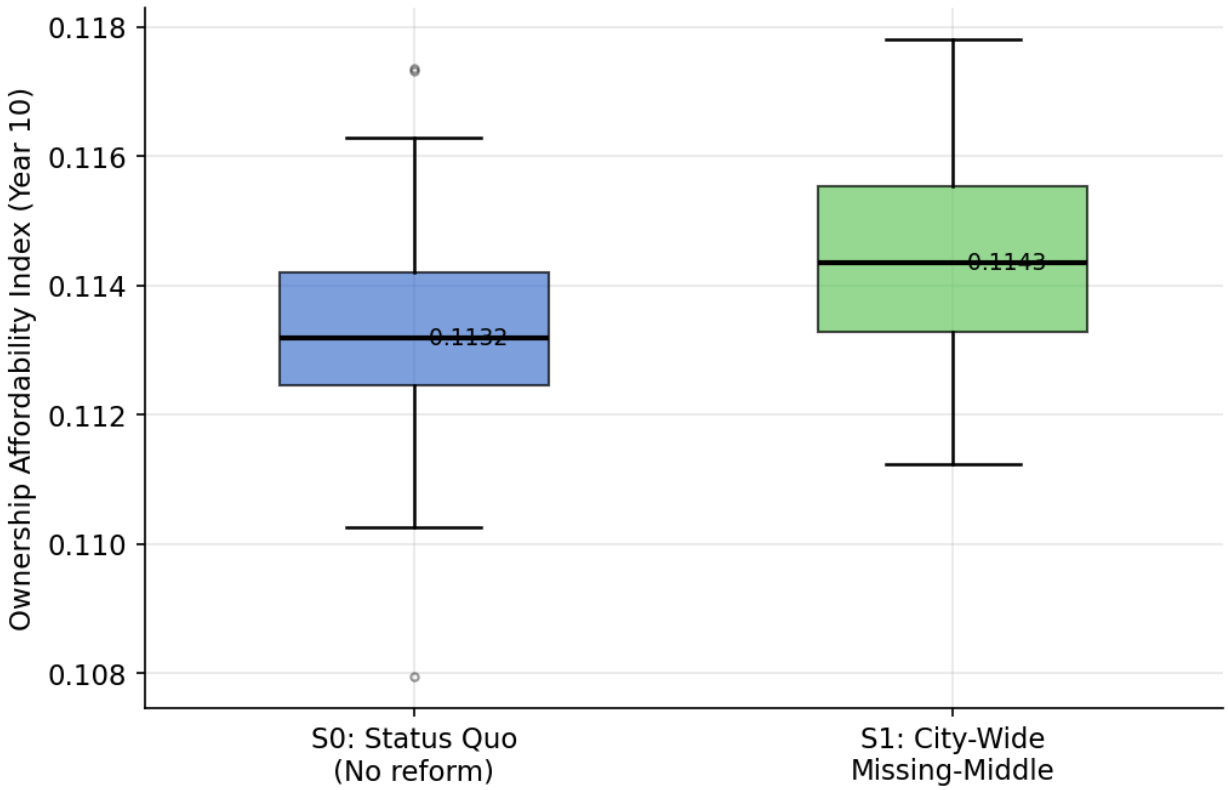

Figure 3: Baseline Scenario (S0) vs City-Wide Zoning Reform (S1) Ownership Affordability at year 10 (N=100 realizations)

Under S1, the city-wide zoning reform enables development across all 1,220 CTs, adding approximately 4,300 units per year on average. The median ownership affordability index at year 10 is 0.1143 under S1 compared to 0.1132 under S0, an improvement of approximately 1%. The S1 distribution is consistently shifted upward relative to S0, with no realization falling below the S0 median, indicating that city-wide missing-middle zoning robustly improves ownership affordability across the range of market conditions simulated.

The magnitude of the improvement is modest over the 10-year horizon. This is expected given that the simulation adds approximately 43,000 units cumulatively under S1 against a baseline stock of approximately 600,000 units, a supply increase of roughly 7%, which theory and empirical literature suggest produces modest price effects in the short to medium term. The Final Report will extend the horizon and refine the market elasticity parameters to better reflect observed Toronto price dynamics.

5 Plan for Next Steps

5.1 Model Improvements and Remaining Work

Several components of the current model are identified in the report as limitations and will be improved in the Final Report.

First, the Infrastructure Model parameters (e.g., capacity growth rate and load coefficients) are currently placeholder values and will be calibrated using more realistic data.

Second, the Development Model will be improved by replacing or supplementing the current linear regression (Stage 2), which has limited explanatory power, with a more flexible model (e.g., random forest) to better capture variation across census tracts.

Third, the Demand Allocation Model will be extended to include additional factors such as transit proximity in the attractiveness function, rather than relying only on inverse housing price.

Fourth, simplifying assumptions such as a uniform initial vacancy rate across all census tracts will be revisited and refined in the Final Report.

Finally, the price and rent elasticity parameters will be calibrated against observed Toronto housing trends to better reflect real-world dynamics.

5.2 Sensitivity Analysis Plan

A sensitivity analysis will be conducted to evaluate how changes in key model parameters affect outcomes. Specifically, we will vary:

- Price and rent elasticity parameters
- Infrastructure model parameters
- Demand growth rate and stochastic variation

The model will be run multiple times under different parameter combinations, and results will be compared using summary statistics (e.g., median affordability index and variability across realizations). This will help assess the robustness of the model and identify the most influential parameters.

5.3 Division of Responsibilities

Chanuth Weeraratna
Improve the Development Model by testing a more flexible alternative to the current linear regression and selecting the best-performing approach. Implement the sensitivity analysis framework, including running simulations across different parameter values.

Devon Kisob
Calibrate Infrastructure Model parameters (e.g., capacity growth and load coefficients) and price/rent elasticity values using external data. Refine the uniform vacancy rate assumption to better reflect variation across census tracts.

Kevin Kim
Extend the Demand Allocation Model by incorporating additional factors (e.g., transit accessibility) into the attractiveness function. Support sensitivity analysis by analyzing and visualizing results across parameter variations.

References

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